Design and Experimental Characterization of a Capacitive Soil Moisture Sensor Using a 555 Timer Architecture

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***Abstract*—This project explores the design and implementation of a capacitive moisture sensing system for plant monitoring applications. To better understand analog front-end design in embedded sensing systems, a capacitive sensor was developed from first principles to measure soil moisture content in household plants. The system continuously monitors soil conditions and notifies the user when watering is required, helping reduce plant neglect and improve plant survival. The final design was implemented as a printed circuit board (PCB)-based probe and experimentally evaluated under varying soil moisture conditions to characterize its performance and operating range.**

***Index Terms*—capacitive sensing, soil moisture sensor, interdigitated capacitor, 555 timer, PCB sensor**

I. Introduction

Simple resistive moisture sensors are commonly used for soil moisture detection, but they suffer from several significant drawbacks. A resistive sensor operates by driving current through one probe, into the soil, and back through a second probe connected to ground. Because current continuously flows between the electrodes, electrochemical reactions occur at the probe surfaces, leading to corrosion over time. This corrosion gradually degrades sensor accuracy, reliability, and operational lifetime. In addition, prolonged electrochemical activity can alter the local soil chemistry through the introduction of metal ions and other byproducts.

Another disadvantage of resistive sensors is their continuous power consumption. Since current is constantly conducted through the soil, the sensor continuously draws energy from the power supply during operation. Capacitive moisture sensors avoid many of these issues. Rather than relying on direct current flow through the soil, capacitive sensors detect changes in the dielectric properties of the surrounding material. Charge is redistributed within the sensing circuit, but there is no continuous conductive current path through the soil to ground. As a result, capacitive sensors experience significantly less electrode degradation and generally consume less power, making them more suitable for long-term sensing applications.

This paper introduces the MSV1 (Moisture Sensor Version 1), a real-time capacitive soil moisture sensing probe designed for household plant monitoring applications. The system uses an interdigitated capacitive comb structure coupled to a 555 timer-based frequency sensing architecture in order to convert soil moisture variations into measurable voltage changes. The sensor was designed, simulated, fabricated, and experimentally characterized under varying soil moisture conditions. Experimental results demonstrate the relationship between soil moisture, sensor capacitance, oscillation frequency, and output voltage, while also identifying several limitations and opportunities for future improvement in the sensing architecture.

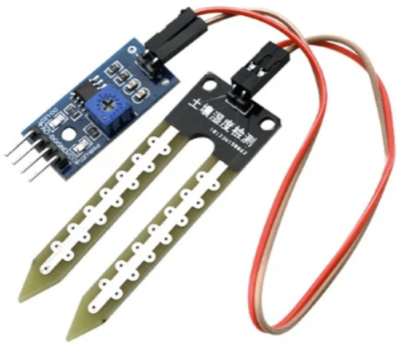

Fig. 1. Resistive moisture sensor. [1]

II. Background/Theory

A. Capacitive Sensing

Capacitive sensing is the process of measuring changes in a target system through variations in capacitance. A change in capacitance alters the RC time constant of the sensing circuit, which in turn changes the frequency of oscillation. Depending on the application, this oscillating signal can then be processed or transformed in several ways to extract useful information about the system being measured. In this application, changes in moisture content affect the effective capacitance of the sensor. As the capacitance changes, the oscillation frequency of the RC network also changes. By monitoring this behavior,

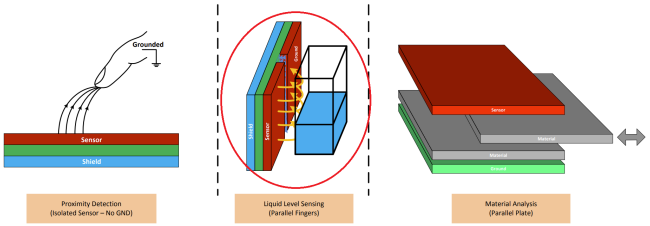

Figure 2-1. Basic Implements for Capacitive Sensing

Fig. 2. Capacitive sensor examples. [2]

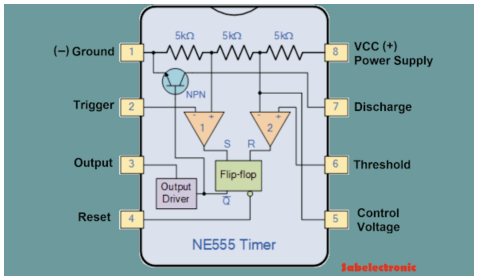

Fig. 3. 555 timer layout [5]

the system can determine relative moisture levels and use this information to indicate when watering is required.

B. 555 timer

The 555 timer is a widely used integrated circuit manufactured by Texas Instruments that can generate precise timing signals and oscillations. In this application, the 555 timer was used to convert a DC supply voltage into a periodic oscillating signal needed for the capacitive sensing circuit. Since the sensor operates by observing changes in the RC time constant of a capacitive network, an oscillating excitation signal was required across the capacitive load. Portable designs are powered by batteries which output a DC voltage, so the 555 is needed to induce these oscillations. The 555 timer operates using internal comparators, a flip-flop, and a discharge transistor. When the trigger input falls below approximately one-third of the supply voltage, the internal comparator sets the output high and allows the external timing capacitor to charge through a resistor. As the capacitor voltage increases, it is monitored by the threshold comparator. Once the capacitor voltage reaches approximately two-thirds of the supply voltage, the output switches low and the internal discharge transistor rapidly discharges the capacitor, returning the circuit to its initial state. Different configurations of the eight pins on the 555 timer allow it to operate in several modes, including monostable, bistable, and astable operation. In this design, the first 555 timer was configured in astable mode, where the external RC network continuously charged and discharged to generate a repeating oscillating waveform. This oscillation was then used to drive the capacitive sensing stage.

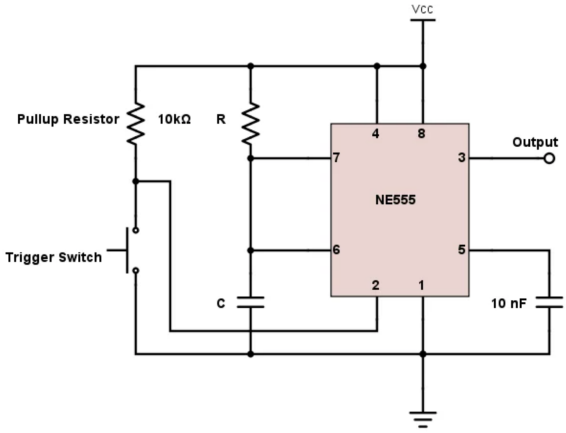

Fig. 4. Example of monostable 555 implementation [4]

C. Monostable 555

Fixed pulse generation using a 555 timer is commonly implemented using the monostable operating mode. As shown in Figure 4, in this configuration, the 555 timer produces a single output pulse of fixed duration whenever it receives a trigger signal. The width of the pulse is determined by an external resistor-capacitor (RC) network connected to the timer. This configuration is different from the sensing 555 timer. Input oscillations will trigger fixed pulse outputs which when averaged with an RC LPF, result in a DC voltage proportional to the speed of input oscillations.

D. Low-Pass Filter (LPF)

A low-pass filter (LPF) is a circuit that attenuates high-frequency components of a signal while allowing low-frequency components to pass through. When designed with a sufficiently large time constant, an LPF can effectively average an AC or pulsed waveform into a DC voltage level. In this sensor, an RC low-pass filter was used to extract a DC signal from the fixed-width pulse output of the monostable 555 timer. Each time the monostable circuit was triggered, it generated a pulse of constant duration. As the trigger frequency increased, the pulses occurred more frequently, causing the RC filter to average the waveform to a higher DC voltage. Conversely, lower trigger frequencies resulted in a lower average DC output voltage. By converting pulse frequency into a proportional DC voltage, the LPF enabled the system to measure changes in the sensor oscillation frequency and use this information to determine moisture levels.

E. Hysteresis

Hysteresis is the process of applying feedback from the output of a comparator circuit back to its input in order to prevent thrashing in sensing applications. The feedback network slightly shifts the switching threshold depending on the current output state. This creates two separate threshold points for rising and falling signals, improving stability near the transition region. In this application, the comparator input

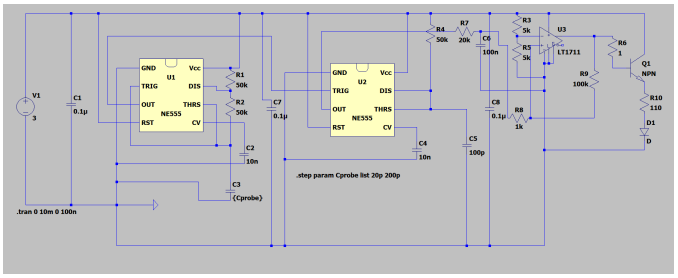

Fig. 5. LTSpice moisture sensor schematic.

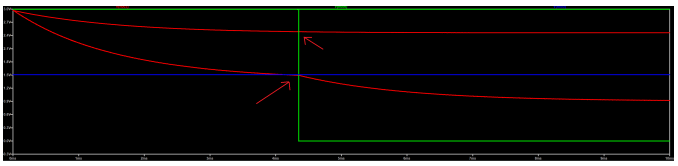

Fig. 6. LTSpice simulation demonstrating hysteresis behavior in the comparator circuit.

is driven by an oscillating RC low-pass filter, which introduces small voltage ripples at the input node. Without hysteresis, these ripples could cause the output to rapidly switch back and forth when the signal is near the threshold voltage. To mitigate this effect, the hysteresis network slightly raises or lowers the effective threshold after each state transition, ensuring stable operation and preventing unwanted oscillation at the switching point. As can be seen in Figure 6, the first red curve on the top represents a capacitance that is lower than desired water content for the plant. This signal is passed through the comparator without any effect. the bottom red curve hits the threshold that the user set indicating that the moisture level is sufficient, and is then pushed even further downward than it would naturally due to hysteresis.

III. Sensor Design

A. Capacitance estimation

TABLE I
Interdigitated Capacitive Sensor Parameters for MSV1

Param.	Description	Value
W	Electrode finger width	0.5 mm
G	Gap between adjacent fingers	0.475 mm
λ	Finger pitch	2.45 mm
L	Electrode finger length	40 mm
N	Number of fingers	75

TABLE II
Approximate Relative Permittivity of Materials [2]

Material	Relative Permittivity (ε_r)
Air	≈ 1
FR4 PCB Substrate	≈ 4.2
Dry Soil	$2-5$
Moist Soil	$10-30+$
Water	≈ 80

The capacitance of the interdigitated sensing structure was estimated using the proportional interdigital capacitor (IDC) relationship described by Igreja and Dias [3]

$$C \approx \varepsilon_0 \varepsilon_{\text{eff}} \frac{NLW}{G}$$

where ε_{eff} is the effective dielectric constant. This value can be roughly estimated with the equation

$$\varepsilon_{\text{eff}} \approx 0.3\varepsilon_{r,\text{air}} + 0.4\varepsilon_{r,\text{FR4}} + 0.5\varepsilon_{r,\text{soil}}$$

and the capacitance can then be estimated with the following equation.

$$\boxed{C \approx (8.854 \times 10^{-12})(\varepsilon_{\text{eff}}) \frac{(75)(0.040)(0.0005)}{0.000475}}$$

Based on the approximate values of ε_{r} from Table II this gives an estimated range of

$$80 \text{ pF} \leq C \leq 220 \text{ pF}$$

for different soil conditions. These approximations are very rough so an empirical analysis was needed to confirm the range. This range was used to select the resistance in the RC sensing oscillator, as the system is only sensitive if the RC constant is within a valid range. Too large or too small and the monostable 555 and the LPF would not function properly converting AC to DC.

B. System Bring-up and Verification

A capacitive probe placed in soil experiences changes in capacitance as moisture levels vary. This capacitance forms part of an RC network that controls the frequency of a 555 timer oscillator. As moisture changes, the oscillation frequency shifts accordingly. This frequency is then processed using a second 555 timer configured to generate fixed-width pulses. These pulses are passed through an RC low-pass filter, producing a DC voltage proportional to the oscillation frequency. The resulting voltage is compared against a user-defined threshold set by a potentiometer using a comparator. When the voltage indicates the soil is too dry, the comparator output drives an NPN transistor, turning on a red LED as an indicator. When moisture is sufficient, the LED remains off.

The MSV1 was initially designed using the native schematic editor in Altium Designer. However, due to high-frequency convergence issues encountered within Altium's simulation environment, LTSpice was used to verify circuit operation and analyze system behavior. The LTSpice simulations were primarily used to determine appropriate resistor values for both the astable 555 oscillator and the monostable 555 timer stages in order to achieve proper operation across the expected sensing capacitance range.

To verify the functionality of the sensing system, a capacitance sweep was performed across the anticipated operating range of the sensor. During the sweep, the output voltage at the positive comparator terminal was monitored and analyzed. The

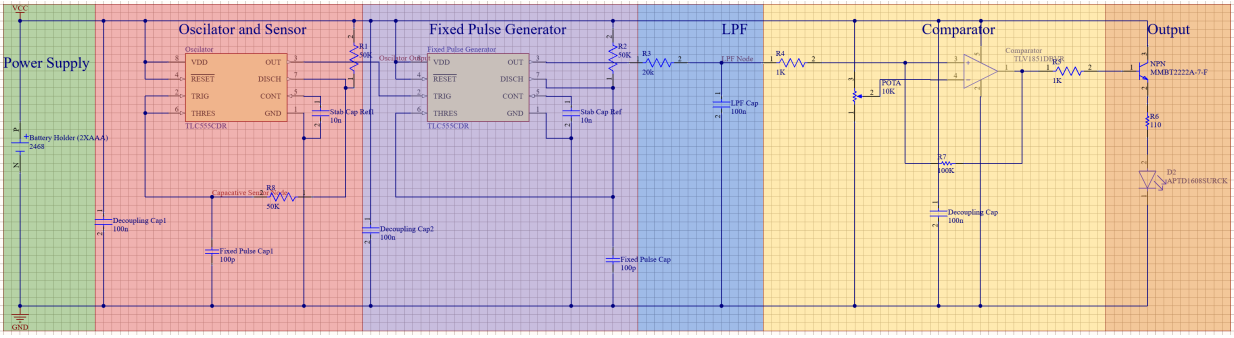

Fig. 7. Altium Designer moisture sensor schematic.

simulation results confirmed that changes in sensing capacitance produced a substantial voltage swing at the comparator input node.

An earlier iteration of the design, which did not include the monostable 555 timer stage, produced only a small voltage variation on the order of a few hundred millivolts. By incorporating the monostable pulse-generation stage, the resulting output swing increased significantly to greater than 2 V, substantially improving the sensitivity and measurability of the sensing system.

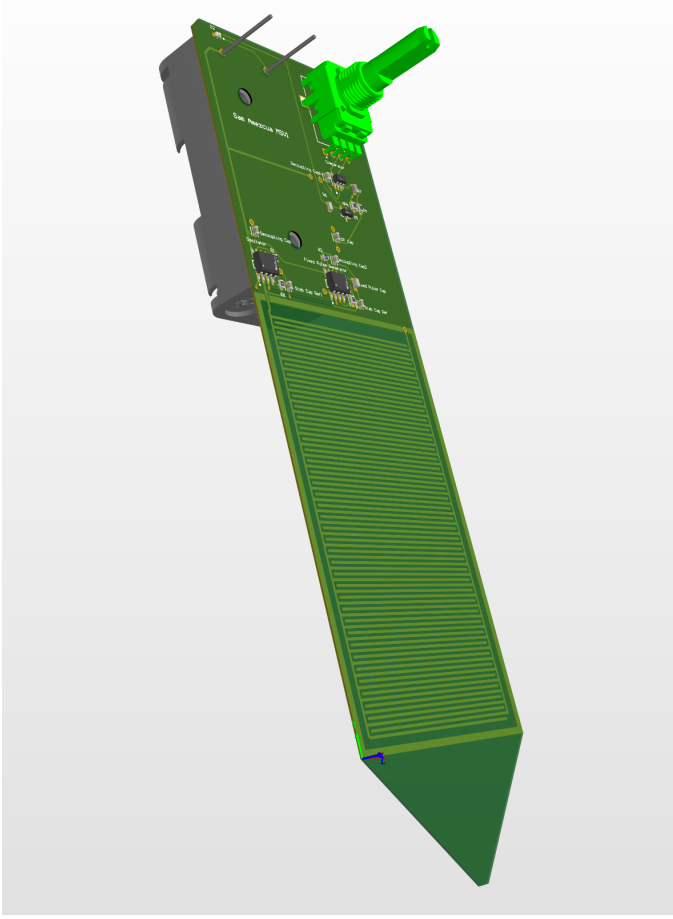

Fig. 8. 3D Altium Designer Moisture Sensor PCB

After verification in LTSpice, the completed schematic was transferred into the Altium Designer PCB layout environment. The capacitive sensing structure consisted of two interdigitated comb-shaped electrode networks. The left electrode comb served as the sensing node connected to the input of the first 555 oscillator stage, while the opposing comb was connected to ground. This interdigitated geometry allowed fringing electric fields to extend into the surrounding soil, enabling changes in moisture content to be detected through variations in capacitance.

To minimize the overall footprint of the design, the AAA battery pack was mounted on the back side of the PCB while the remaining circuit components were placed on the front side. The objective of this prototype was to provide a simple physical verification of the sensing system and demonstrate successful moisture detection functionality.

A small 0603 red LED was positioned near the top of the probe to provide a visual indication when additional watering was required. Once the PCB layout was completed, the design files were submitted to JLCPCB for fabrication. The manufactured prototype was delivered approximately one month after submission for review.

IV. Experimental Results and Analysis

Upon receiving the finished product from JLCPCB, some basic assembly and soldering were required to bring the board into working condition. The potentiometer and AAA battery holder were through-hole components that JLCPCB did not populate. After these components were soldered onto the PCB, the basic functionality of the sensor was verified. The sensor successfully turned the LED on and off based on the surrounding moisture conditions. Human hands were used to approximate high-moisture conditions, while ambient air represented low-moisture conditions. During testing, an issue arose in which the sensor would become “stuck” in one of its output states. Once the moisture conditions caused the circuit to switch to the LED-off state, it could not reliably return to the LED-on state, which was intended to indicate that watering was required.

By analyzing the voltages at the positive and negative comparator terminals, it became clear that the hysteresis network was the source of the problem. Hysteresis had been included

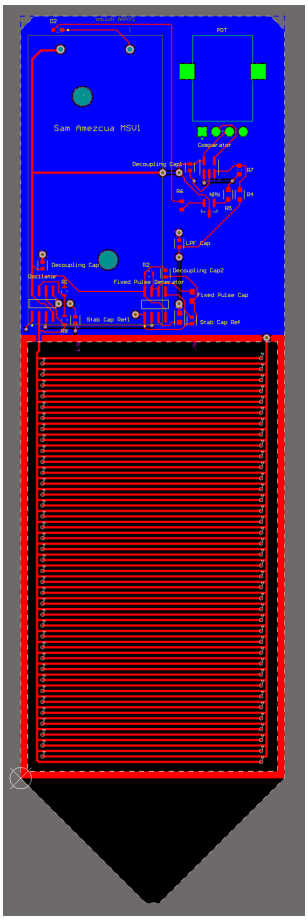

Fig. 9. 2D top-layer Altium Designer Moisture Sensor PCB

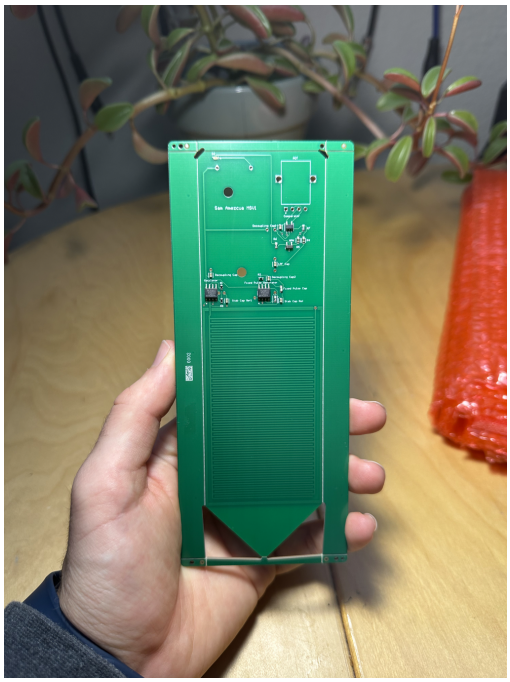

Fig. 10. JLCPCB Manufactured MSV1 PCB

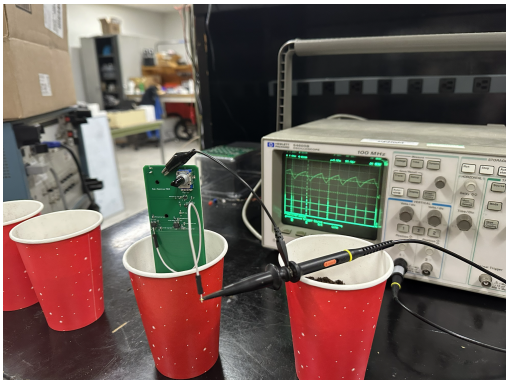

Fig. 11. Experiment 1 Setup

to prevent rapid switching when the input voltage was near the threshold value. The hysteresis network consisted of a 100 kΩ feedback resistor and a 1 kΩ series resistor connected to the comparator input. However, the 100 kΩ feedback resistor was too small for this sensing application, causing the feedback voltage to shift the threshold excessively high or low. As a result, the circuit became latched in either state.

To remedy the instability issues observed during initial testing, resistor (R_7), the 100 kΩ feedback resistor, was removed from the PCB in order to eliminate the feedback path entirely. Removing this resistor immediately resolved the latching behavior present in the system and allowed the circuit to reliably switch between low and high-capacitance states. These transitions were now accurately reflected through the LED indicator output.

Another modification made to one of the five prototype PCBs received from JLCPCB involved the addition of external probe points to facilitate easier measurements of critical signals within the circuit. Two wires were soldered directly onto the PCB. The first wire was attached to the capacitive sensing node of the comb electrode structure in order to measure the oscillation frequency and derive the effective capacitance of the sensor. The second wire was attached at the input of the comparator to measure the output voltage of the low-pass filter. These additional measurement points enabled accurate logging and characterization of the system under varying moisture conditions.

After verifying proper operation of the modified system, a series of experiments were conducted to validate the sensing behavior and compare the measured results against both simulation predictions and rough analytical capacitance estimates.

A. Experiment 1

The first experiment examined the response of the sensor to varying moisture levels across two different soil types. Two categories of soil were selected: a dense potting mix and a lighter cactus mix. Three cups of each soil type were prepared, and each cup was filled with a different quantity of water ranging from $\frac{1}{3}$ cup to 1 cup. In hindsight, the highest moisture level was excessive, as the 1-cup condition effectively

TABLE III
FREQUENCY, VOLTAGE, AND DERIVED CAPACITANCE VERSUS SOIL MOISTURE CONTENT

Condition	Potting Soil				Cactus Soil			
	Dry Soil	1/3 Cup	1/2 Cup	1 Cup	Dry Soil	1/3 Cup	1/2 Cup	1 Cup
Frequency	27 kHz	15 kHz	10 kHz	7.5 kHz	25 kHz	19.5 kHz	11.3 kHz	7.7 kHz
Voltage	1.10 V	1.06 V	1.08 V	1.08 V	1.47 V	1.08 V	1.08 V	1.07 V
Calculated Capacitance	356 pF	640 pF	960 pF	1.28 nF	384 pF	492 pF	850 pF	1.25 nF

Control (Dry Air)	Value
Frequency	35 kHz
Voltage	1.56 V
Capacitance	274 pF

produced a suspension of soil floating in water rather than realistic moist soil conditions.

The capacitive probe was inserted into each sample, and measurements were taken at two locations within the circuit. The first measurement was the oscillation frequency at the capacitive sensing node, while the second was the output voltage at the comparator stage. The resulting data are summarized in Table III.

The results show that the oscillation frequency at the capacitive sensing node decreases significantly as the moisture content increases. This behavior is expected, since increasing soil moisture raises the effective dielectric constant surrounding the capacitive comb structure, thereby increasing the capacitance of the sensor. Because the oscillation frequency of the 555 timer is inversely proportional to capacitance, the frequency decreases as moisture increases.

1) Capacitance Derivation from Oscillation Frequency: According to the standard astable configuration equation for the 555 timer [6], the oscillation frequency is given by

$$f = \frac{1.44}{(R_A + 2R_B)C}$$

where R_A and R_B are the timing resistors and C is the effective capacitance connected to the threshold and trigger nodes of the timer.

For this design,

$$R_A = 50\,k\Omega$$

$$R_B = 50\,k\Omega$$

which produces

$$R_A + 2R_B = 150\,k\Omega$$

Substituting these values into the astable oscillator equation gives

$$f = \frac{1.44}{150 \times 10^3 \times C}$$

Solving for capacitance yields

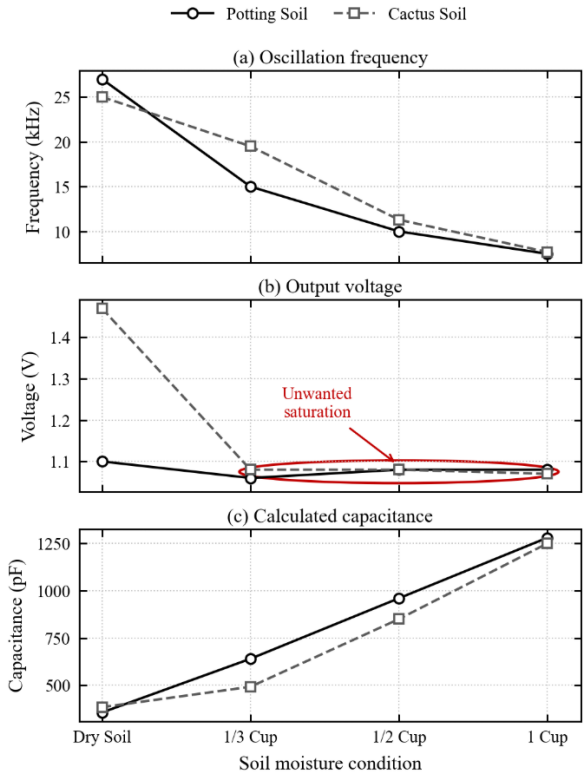

Fig. 12. Experiment 1 Frequency, Voltage, and Derived Capacitance Versus Soil Moisture Content

$$\boxed{C = \frac{1.44}{150 \times 10^3 \times f}}$$

Using the experimentally measured oscillation frequencies, the effective capacitance of the sensing node was calculated for each moisture condition. The resulting capacitance values ranged from approximately 270 pF in dry air to approximately 1.2 nF in highly saturated soil conditions.

These experimentally derived capacitance values were significantly larger than the originally estimated range of 80 pF to 220 pF predicted during the design phase. This discrepancy likely contributed to the saturation behavior observed at the

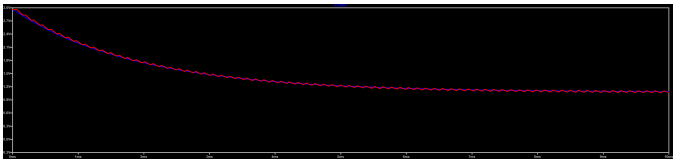

Fig. 13. LTSpice Simulation of 270pF Sensor Capacitance (blue) and 1.2nF Sensor Capacitance (red)

voltage output of the system. The timing components of the sensing circuit were originally selected such that the system would exhibit maximum sensitivity within approximately the 100 pF operating region. However, the measured capacitances reached values nearly an order of magnitude larger under highly saturated soil conditions.

This behavior was further verified in LTSpice simulation in Figure 13 by selecting sensor capacitance values of 274 pF and 1.2 nF, corresponding approximately to the experimentally measured dry-air and highly saturated soil conditions, respectively. Despite the nearly order-of-magnitude difference in capacitance, the resulting output voltage waveforms converged toward the same steady-state value.

These simulation results strongly support the experimental observation that the circuit was not optimally tuned for operation within the actual capacitance range produced by the interdigitated sensor structure. The timing constants of the astable oscillator, monostable pulse generator, and low-pass filter were originally selected based on significantly lower estimated capacitance values. As the sensor capacitance increased beyond the intended operating range, the system gradually lost sensitivity and entered a saturation region where further capacitance changes produced only minimal output voltage variation.

This analysis suggests that substantial improvements could be achieved in future iterations of the design by re-scaling both the sensor geometry and the RC timing networks around the experimentally measured capacitance range. With properly selected component values, the system should maintain a significantly larger voltage swing across realistic soil moisture conditions, thereby improving overall sensor sensitivity.

B. Experiment 2

The second experiment involved probing the outputs of both the astable 555 oscillator and the monostable 555 oscillator in order to determine whether the fixed-width pulse generator stage was necessary for proper operation and increased sensing functionality. Moisture variation was introduced manually by wrapping human hands around the capacitive probe, thereby increasing the effective capacitance through the dielectric properties of the human body. Measurements were taken at both timer outputs to compare waveform characteristics and observe how the fixed pulse shaping stage influenced the final DC output voltage after filtering. This experiment helped verify the functional purpose of the monostable stage and determine whether a simpler architecture could potentially achieve similar sensing behavior.

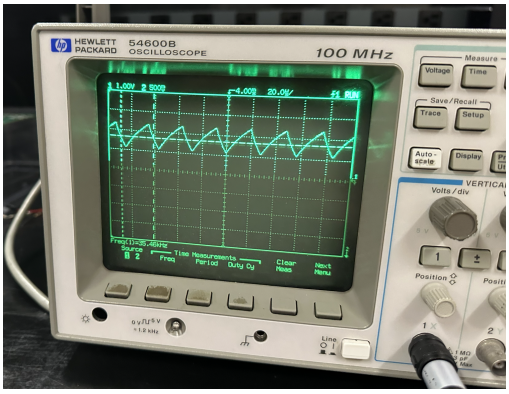

Fig. 14. Capacitive Sensing Node Waveform

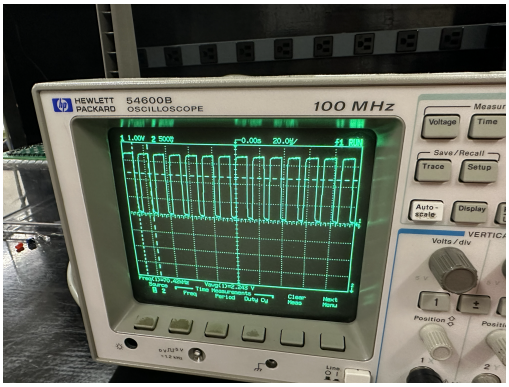

Fig. 15. 555 Astable Oscillator Output Waveform

It was observed experimentally that the measured frequency at the astable output was approximately twice the frequency measured directly at the capacitive sensing node. This discrepancy is likely due to differences in waveform shape and oscilloscope triggering behavior between the analog RC timing waveform and the digital square-wave output of the 555 timer. Despite the frequency measurement difference, both signals responded consistently to changing capacitance and confirmed proper oscillator operation.

From these results, it can be concluded that the addition of the monostable 555 timer stage was an effective design choice for the sensing circuit. Without the monostable pulse-shaping stage, the frequency changes produced by the capacitive oscillator were largely ignored after averaging through the low-pass filter. This behavior is theoretically expected because the astable oscillator primarily changes in frequency while maintaining an approximately constant duty cycle. Although the oscillation period changes as the capacitance varies, the relative proportion of time the waveform spends at (V_{CC}) versus ground remains nearly constant. As a result, the average DC value of the oscillator waveform changes very little even for large shifts in oscillation frequency.

The experimental results strongly support this analysis. The direct oscillator output exhibited less than a 100 mV change despite a frequency reduction of approximately 29 kHz, de-

TABLE IV
OSCILLATOR AND FIXED PULSE OUTPUT RESPONSE TO HAND PROXIMITY

Measurement	Air	Hands on Probe
Average Oscillator Output Voltage	2.2 V	2.2 V
Oscillator Output Frequency	69 kHz	40 kHz
Average Fixed Pulse Output Voltage	1.55 V	1.08 V
Fixed Pulse Output Frequency	69 kHz	40 kHz

Fig. 16. Experiment 3 Setup, Multimeter

creasing from 69 kHz to 40 kHz. In contrast, the monostable output stage produced a voltage drop of approximately 500 mV over the same frequency range. This demonstrates that the fixed-width pulse generator effectively converts frequency variation into a significantly larger change in average DC voltage, thereby greatly improving the sensitivity of the sensing system.

These results are highly encouraging and validate the overall architecture of the design. Furthermore, the sensitivity of the system could likely be improved even further in future iterations by incorporating the capacitance characterization insights obtained from the previous experiment. By redesigning the timing constants of the oscillator and monostable stages around the experimentally measured capacitance range, the output voltage swing could be increased substantially, resulting in improved moisture resolution and overall sensor performance.

C. Experiment 3

The final experiment evaluated the long-term sensing capability of the system. The capacitive probe was inserted into the cactus soil sample containing $\frac{1}{3}$ cup of water and allowed to remain in place over the course of several hours. A household fan was turned on above the experiment to speed up the process of soil drying. Measurements were periodically recorded every 30 min in order to determine whether the sensor

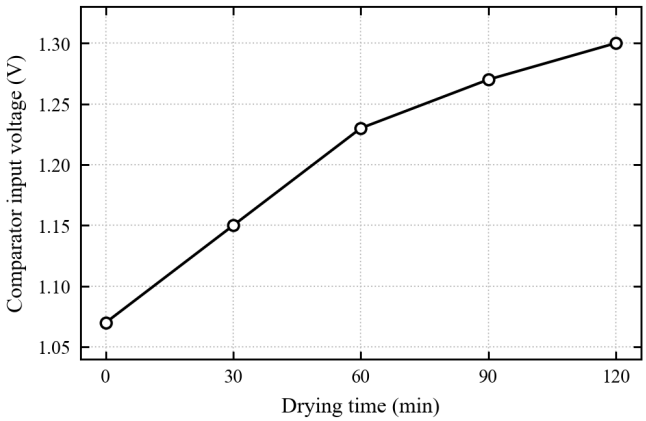

Fig. 17. Experiment 3, voltage at input of comparator versus time

could accurately track the gradual reduction of moisture within the soil as evaporation occurred.

This experiment was the most important validation of the overall project because it directly evaluated the intended real-world functionality of the sensor. Rather than detecting only large instantaneous changes in moisture, the system was required to demonstrate the ability to monitor gradual environmental changes over extended periods of time. The following results summarize the observed long-term behavior of the sensor.

As shown in Figure 17, the voltage at the comparator input increased steadily as the soil dried over time. Eventually, the voltage crossed the comparator threshold and the LED indicator illuminated, demonstrating that the sensor was capable of detecting gradual moisture loss under realistic operating conditions.

However, instability was observed near the switching threshold. After the removal of resistor R_7, the comparator no longer possessed hysteresis functionality. As a result, small voltage fluctuations near the threshold caused the LED to flicker on and off intermittently for approximately 10 minutes during the transition region. This behavior is undesirable for a practical sensing system because it introduces ambiguity near the watering threshold.

This issue could likely be corrected in future iterations by reintroducing the hysteresis network using a substantially larger feedback resistor value, potentially on the order of 100 MΩ or greater, in place of the original 100 kΩ resistor. A larger resistance would preserve the stabilizing benefits of hysteresis while minimizing the excessive threshold shifting that caused the original latching behavior observed in the MSV1 prototype.

V. CONCLUSION AND FUTURE WORK

The MSV1 successfully detected capacitance changes resulting from varying soil moisture conditions. Experimental results verified the effectiveness of the astable–monostable 555 timer architecture and demonstrated that the addition of the monostable pulse-shaping stage significantly improved

system sensitivity. The interdigitated capacitive comb structure successfully converted moisture variations into measurable frequency and voltage changes.

Experimentally derived capacitance values were significantly larger than the originally predicted range, explaining the voltage saturation behavior observed during testing and simulation. In addition, the hysteresis feedback network negatively impacted system functionality by excessively latching the comparator output, requiring removal of the feedback resistor for proper operation.

These results provide a clear direction for future improvements. Future versions of the sensor can be redesigned around the experimentally measured capacitance range by resizing the interdigitated comb structure and retuning the RC timing networks of the astable and monostable stages. In particular, reducing the finger length of the sensor would decrease the overall capacitance range and improve sensitivity within realistic soil moisture conditions.

The physical dimensions of the system could also be reduced substantially through the use of smaller components and a more compact sensing geometry. Additionally, the hysteresis network could be reintroduced with optimized resistor values to provide noise immunity without introducing excessive latching behavior. Future iterations would also benefit from a protective enclosure for long-term deployment and integration with a microcontroller-based automatic watering system.

Overall, this project served as a valuable introduction to capacitive sensing, analog timing circuitry, PCB design, and the complete engineering process from simulation to fabrication and experimental characterization. The analysis performed on the MSV1 prototype provides a strong foundation for the development of future sensor revisions.

ACKNOWLEDGMENT

I would like to thank my dog Eddie for keeping me company during the many long nights spent working on this project, and my fiancée for patiently listening to far more discussion about moisture sensors than anyone should reasonably have to endure. I truly could not have completed this project without their support. Additional thanks go to my mom for providing the potting soil used throughout the experimental testing.

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